$$\int e^{at}\underbrace{(\cos(bt)+i\sin(bt))}_{e^{ibt}}dt = \int e^{(a+ib)t} = \frac{1}{a+ib}e^{a+ibt}+C$$

$$= \frac{a-ib}{a^2+b^2}e^{at}\cdot(\cos(bt)+i\sin(bt))+C$$

$$= e^{at}\cdot\left(\frac{a}{a^2+b^2}\cos(bt)+\frac{b}{a^2+b^2}\sin(bt) + \frac{a\sin(bt)}{a^2+b^2}i - \frac{b\cos(bt)}{a^2+b^2}i\right)$$

$$= e^{at}\left(\frac{a\cos(bt)+b\sin(bt)}{a^2+b^2} + \frac{a\sin(bt)-b\cos(bt)}{a^2+b^2}i\right)+C$$

Notice that, $\int e^{at}(\cos(bt)+i\sin(bt))\,dt = \int e^{at}\cos(bt)dt + i\int e^{at}\sin(bt) = G_{\cos}(t)+iG_{\sin}(t)+C$

$$\Rightarrow \int e^{at}\cos(bt)\,dt = e^{at}\cdot\frac{a\cos(bt)+b\sin(bt)}{a^2+b^2}+C, \quad \int e^{at}\sin(bt)\,dt = e^{at}\cdot\frac{a\sin(bt)-b\cos(bt)}{a^2+b^2}$$

Let $\vec{c} = \begin{pmatrix} a \\ b \end{pmatrix}$, $\vec{v} = \begin{pmatrix} \cos(bt) \\ \sin(bt) \end{pmatrix}$

$$\int e^{at}\cos(bt)\,dt = e^{at}\cdot\frac{\vec{c}\cdot\vec{v}}{\|\vec{c}\|^2}, \quad \int e^{at}\sin(bt)\,dt = e^{at}\cdot\frac{\vec{c}\times\vec{b}}{\|\vec{c}\|^2}$$